# Thesis proposal on Parliamentary Studies

Bocconi University | Department of Social and Political Science

Parliamentary Studies

**Filippo Panfoli 3158433**

filippo.panfoli@studbocconi.it | PanfoliF |
0009-0000-7861-8036 |

**LaTeX sources:** PanfoliF/

**Version:** v1.1 | **Last updated:** April 23, 2026

**Prepared for:** Parliamentary Studies, taught by Prof. Morelli

**License:** CC BY 4.0

April 23, 2026

## Abstract

This article explores the topic of linguistic simplification in parliaments. In 1868 a law broadened the electorate base of UK House of Commons. Can we exploit this setting to measure the effects on linguistic simpilification? Spirling (2016) proved that a broader electoral base has different effects on linguistic simpilification for cabinet and non cabinet members. Building on Spirling (2016), I try to support the causal inference analysis with further investigation of data. In particular, I disentagle the effects of to different measures on the outcome variable provided by Spirling (2016). I will also reduce the time window to empower the effects of observations around 1868 cutoff.

Text as data techinques are used the selct the best method to get a DTM matrix out of the raw speeches. Causal Inference methods are employed on this data. After these steps, I lay down the data generating process, I use a causal diaciclycal graph to build the causal mechanism. These sections should be a suitable theoretical ground for the application of an regression discontinuity design.

# 1 Introducation

This second part of the paper is dedicated to test the analysis conducted by Professor Atrhur Spirling (Spirling, 2016). Actually, it provides good reasons to not stop doing research on the topic. In fact, it does not offer any decisive or comprehensive proof of any hypothesis. It suggest that slighlty moving the focus of regressions and inferece closer to the data of 1868 makes us change perpective on linguistic simplification.

*Knowledge of Language:* A limit of the present proposal is linguistic knowledge. In order to strenghthen the results of causal inference it is good to have deep understanding of the language used at the time. This would also shape the selection of n-grams, discriding irrelevant term and valuing relevant expressions.

*Data:* When I first tried to reproduce Spirling (2016)'s replication package I found out that it is incomplete. I texed Professor Spirling (2016) and he confirmed that the first part of the pipeline is missing.

The first aim of the analysis is to produce that first part of the pipeline and test whether the results lead to the same conclusion. I already retrieved data from the UK Parliament website and I have already computed the basic linguistic measures.

*DGP and Theoretical Model:* I intend to lay down a game theory model to describe and formalise the structure of the debate inside the Parliament. I will follow the most relevant examples in the literature such as Bäck and Debus (2016) and Proksch and Slapin (2014). The theoretical model will provide the basis for the Data Generatig Process.

*Causal Inference:* In this section I present the intuition that challenges the results of Spirling (2016). According to Professor Spirling analysis, after the broadening of electorate base of UK's House of Commons the language used in parliamentary speeches suffered of simplification in terms of lenghth of sentences and words used. In particular, Spirling (2016) tests the hypothesis that the treatment affects cabinet members language more than non-cabinet members language.

In the first part of the causal inference section I will challenge the hypothesis that the treatment affects at all the language in the parliament. I will use a DinD to estimate the coefficient of the the difference between linguistic trend in the two housesof the UK Parliament. Since the house of Lords is a non-elective chamber, the widening of the electorate should not influence the linguistic measures of the debate in this chamber. Thus, the measures of this house should provide a good control group (if parallel trend assumption holds).

In the second part I will build a topic model to create variables that describe to what extent a debate in parliament belongs to a specif topic. I will use these variables to understand how different topics influence the dependent variable.

In the third part I narrow down the window of data and I challenge Spirling (2016)'s main hypothesis, namely the treatment affects cabinet members language

more than non-cabinet members language. I still have to find support for this test. I will deep my knowledge on bandwidth selection to understand the main drivers of this choice and its main effects on the resulting coefficients.

In the fourth part I will use topic model to understand whether the hypothesis to topic specification.

While in the original papaer from Spirling (2016), they use only one linguistic measure. I intend to provide the results from more statistics, to avoid biases that may arise from a single measure.

# 2 Causal Inference

## 2.1 RDD

The identification strategy emplyed in this proposal is regression discontinuity design. Among the classical literature on RDD, I like highlighting two great contributions: Fujiwara (2015) and Gulzar and Pasquale (2017). A more recent study on RDD methodology is presented by Cattaneo et al. (2024).

The equation used is the following:

$$FRE = \beta_0 + \beta_1 \cdot Cutoff + \gamma X + \varepsilon$$

$$FRE = \beta_0 + \beta_1 \cdot C + \gamma X + \varepsilon$$

Identification strategy relies on:

- $C$ is the treatment. It is an exgoneous variable independet from the others called cutoff. In our regression the cutoff is 1868, namely the year in which the reform was introduced
- $Y$ is any of the measures of linguistic simplification employed.

*Controls:* The vector of controls is $X$, while $\gamma$ is the vector of coefficients. It represents the effects of control variables on our outcome variable. The selected controls are:

- Competitiveness (of electoral distric)
- Party

These variables are the same as Spirling (2016).

## 2.2 DinD Approach

Spirling (2016) uses another approach, Difference in Difference, to test the effects electorate reform on linguistic simplification. This design builds on RDD. However, to safely apply it we require another assumption, *parallel trends assumption.* Parallel Trends Assumption states that absent the treatment, control group and treated group would have

had parallel trends. However, It is impossible to test it empirically. Therefore, it requires qualitative support. The equation of DinD is:

$$FRE = \beta_0 + \beta_1 \cdot Cutoff + \beta_2 \cdot Treatment + \gamma X + \varepsilon$$

$$FRE = \beta_0 + \beta_1 \cdot C + \beta_2 \cdot Cabinet + \gamma X + \varepsilon$$

In this setting cabinet status is a dummy variable that represents the treatment.

Identification strategy adopted by Spirling (2016) relies on the same variables as before:

- $C$ is the treatment. It is an exgoneous variable independet from the others called cutoff. In our regression the cutoff is 1868, namely the year in which the reform was introduced
- $Y$ is any of the measures of linguistic simplification employed.

*Controls:* The vector of controls is $X$, while $\gamma$ is the vector of coefficients. It represents the effects of control variables on our outcome variable. The selected controls are:

- Competitiveness (of electoral distric)
- Party

These variables are the same as Spirling (2016).

## 2.3 DAG

Steiner et al. (2017) propose a causal dyciclical graph for RDDs. This is the typical design for this causal inference technique, and it is the design emplyed in this proposal. In particular:

- $U$ stands for unobservable variable.
- $X$ represents speeches made in Parliament by a member
- $D$ is the treatment, thus beign or not beign a cabinet member in a session of the Parliament.
- $c_0$ is an exgoneous variable independet from the others called cutoff.
- $Y$ is any measure of linguistic simplification.

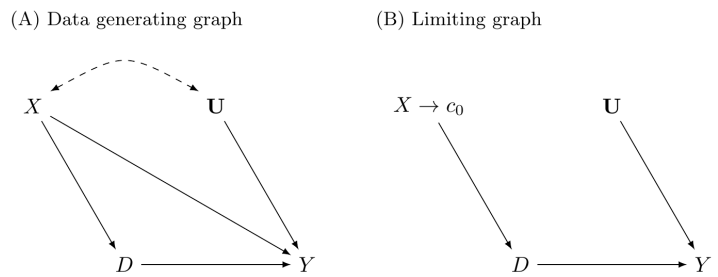


**Figura 1:** *DAG for Regression Discontinuity Design Steiner et al. (2017)*

The DAG formalises the causal mechanism causal inference relies on. It is a usaful way to identify possible weakness of the identification strategy. It strenghthen the analysis showing how the effects of treatment on outcome variable is isolated from the effects of confounders.

In this setting, variables like "Party" and "Competitivness" are confounders. They provide endogenous sources of variation. We take solve this problem controlling for these variables in our regression.

Omitted variable bias happens when we do not control for all confounders. However, it is impossible to eliminate all sources of endogenous variation because some confounders are not observed. In this instance, the solution is not empirical but qualitative. Therefore, it depends on the knowledge of the context under study.

## 2.4 RDD Results

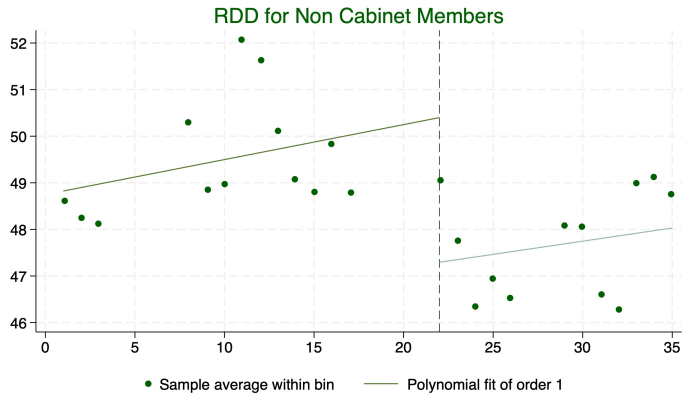


**Figura 2:** *LATE Estimate of RDD for Non Cabinet Members*

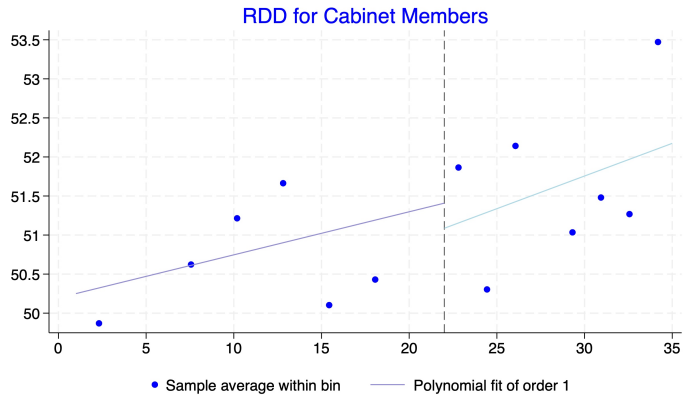


**Figura 3:** *LATE Estimate of RDD for Cabinet Members*

## 2.5 Robustness

### 2.5.1 Bandwidth

Process of selection of optimal window of observation has been extensively studied. Nowadays, most statistical softwares provide command that run optimisation process to automatically select the window.

G. W. Imbens and Angrist (1994) discourges the use of high-order polynomials in the to estame the LATE, they argue that "higher-order polynomials can lead to overfitting and have been found to introduce bias". In the previous pages () we already highlighted the role of overfitting. In particular, we highlighted the contributionof Cranmer and Desmarais (2017).

A way of overcoming the issue of overfitting is using nonparametric kernels (Cunningham, 2021, chap.∽ 6.2.3). However, in this method the selection of bandwidth is crucial. In fact, it "is sensitive to the choice of bandwidth, but more recent work allows the researcher to estimate optimal bandwidths Calonico et al. (2014) and G. Imbens and Kalyanaraman (2012)."

Another issue, namely estimation function, is not stritly related to bandwidth selection. However, Lee and Lemieux (2010) stress the importance of this step. They argue that "allowing different functions on both sides of the discontinuity should be the main results in an RDD paper".

### 2.5.2 Manipulation Tests

In order to strengthen the support to results, I conducted a McCrary Density Test. This test was introduced by McCrary (2008). This test reduces the probability that treated unites "manipulat" the cutoff. In this case it makes us sure of the fact that there is no anticipation effect from Parliament Members. Parliament Members of a party could have decided to rintroduce the reform only when they became sure of having acquired the majority of new voters. This would violate the assumptions that lay at the basics of the regression discontinuity design. The test, conducted for both cabinet and non cabinet, shows that *there is no manipulation.*

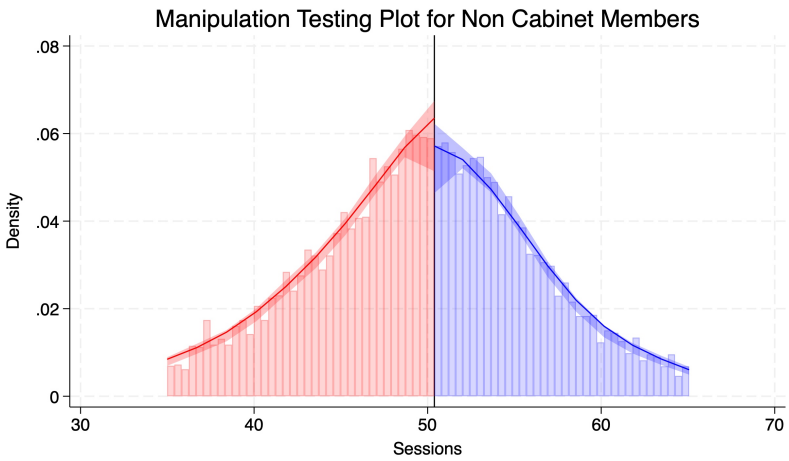


**Figura 4:** *McCrary Manipulation Test for Distribution of Non Cabinet Members around catoff off 1868 election*

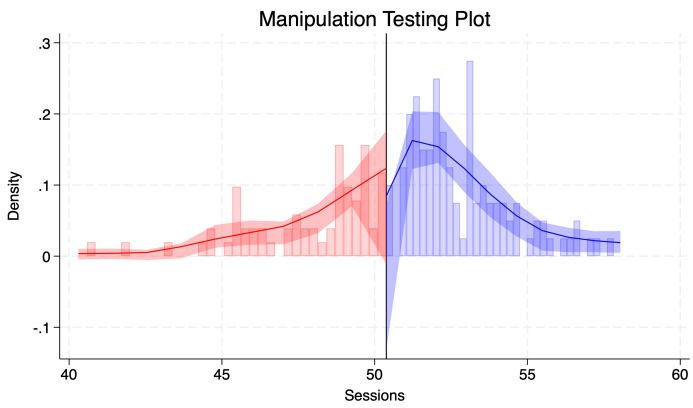


**Figura 5:** *McCrary Manipulation Test for Distribution of Cabinet Members around catoff off 1868 election*

# Bibliography

## Books


Bäck, H., & Debus, M. (2016). *Political Parties, Parliaments and Legislative Speechmaking*. Palgrave Macmillan UK. https://doi.org/10.1057/9781137484550

Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2024, April 30). *A Practical Introduction to Regression Discontinuity Designs: Extensions* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781009441896

Cunningham, S. (2021). *Causal inference: The mixtape*. Yale University Press.

Proksch, S.-O., & Slapin, J. B. (2014, December 4). *The Politics of Parliamentary Debate: Parties, Rebels and Representation* (1st ed.). Cambridge University Press. https://doi.org/10.1017/CBO9781139680752


## Articles


Calonico, S., Cattaneo, M. D., & Titiunik, R. (2014). Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs: Robust Nonparametric Confidence Intervals. *Econometrica*, *82*(6), 2295–2326. https://doi.org/10.3982/ECTA11757

Cranmer, S. J., & Desmarais, B. A. (2017). What Can We Learn from Predictive Modeling? *Political Analysis*, *25*(2), 145–166. https://doi.org/10.1017/pan.2017.3

Fujiwara, T. (2015). Voting Technology, Political Responsiveness, and Infant Health: Evidence From Brazil. *Econometrica*, *83*(2), 423–464. https://doi.org/10.3982/ECTA11520

Gulzar, S., & Pasquale, B. J. (2017). Politicians, Bureaucrats, and Development: Evidence from India. *American Political Science Review*, *111*(1), 162–183. https://doi.org/10.1017/S0003055416000502

Imbens, G., & Kalyanaraman, K. (2012). Optimal Bandwidth Choice for the Regression Discontinuity Estimator. *The Review of Economic Studies*, *79*(3), 933–959. https://doi.org/10.1093/restud/rdr043

Imbens, G. W., & Angrist, J. D. (1994). Identification and Estimation of Local Average Treatment Effects. *Econometrica*, *62*(2), 467. https://doi.org/10.2307/2951620

Lee, D. S., & Lemieux, T. (2010). Regression discontinuity designs in economics. *Journal of Economic Literature*, *48*(2), 281–355. https://doi.org/10.1257/jel.48.2.281

McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics*, *142*(2), 698–714. https://doi.org/10.1016/j.jeconom.2007.05.005

Spirling, A. (2016). Democratization and Linguistic Complexity: The Effect of Franchise Extension on Parliamentary Discourse, 18321915. *The Journal of Politics*, *78*(1), 120–136. https://doi.org/10.1086/683612

Steiner, P. M., Kim, Y., Hall, C. E., & Su, D. (2017). Graphical Models for Quasi-experimental Designs. *Sociological Methods & Research*, *46*(2), 155–188. https://doi.org/10.1177/0049124115582272